

Deepak Mallampati

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PROFESSIONAL SUMMARY

Software engineer (MS CS, 2.5+ years) specializing in production-grade AI systems for regulated clinical and scientific data workflows. Built RAG systems with 90% grounding alignment and agentic workflows for multi-step clinical reasoning - directly applicable to Owkin K's pharmaceutical and biotech deployment environments. HIPAA-conscious data governance (de-identification, data lineage, audit trails) maps directly to pharma/regulatory compliance. Expert in backend AI systems: FastAPI, Docker, Python, LangChain, LlamaIndex. Comfortable working directly with scientific stakeholders (biologists, clinicians, data scientists).

CORE COMPETENCIES

Clinical AI RAG Systems • Agentic Scientific Workflows • Regulated Data Governance (HIPAA) • Backend System Integration • Rapid AI Prototyping • LangChain / LlamaIndex / Vector Search • Python / FastAPI / Docker • Data Pipeline Engineering • Stakeholder Engagement (Technical + Scientific)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting (USA)

- Built **production-grade RAG systems** for clinical knowledge retrieval; recursive semantic chunking, transformer embeddings, reranking; improved precision ~**35%**, grounding alignment exceeded **90%**.
- Developed **agentic workflows** for multi-step clinical queries (eligibility, workflow navigation, documentation); improved stability ~**25%**, hallucinations >30% reduced.
- Built EHR data preprocessing pipelines (Pandas, NumPy); raised dataset reliability above **98%**.
- Maintained **HIPAA-conscious data governance**: de-identification, data lineage, evaluation audit trails - directly analogous to pharma/regulatory compliance requirements.
- Packaged inference as FastAPI/Flask APIs with Docker; integrated into healthcare enterprise systems.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

- Transformer-based hierarchical summarization; RAG document Q&A (reduced hallucinations ~30%, raised accuracy >85%). Flask API for non-technical staff.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- SQL optimization, Jenkins CI/CD, Grafana observability for enterprise production systems.

KEY PROJECTS

Agentic Claims Processing & Fraud Risk Intelligence - Clinical Domain

Multi-agent pipeline with schema-validated JSON contracts between agents. **RAG-grounded regulatory validation** against vector-indexed CPT/ICD policy documents. ANN-based duplicate detection. Explainable risk scoring with audit-ready reasoning traces. Directly analogous to Owkin K's production AI deployment in regulated pharma/biotech environments.

Video Summarization - Transformer Domain Adaptation (Suvidha)

Hierarchical abstractive summarization over 5,000+ sessions; timestamp-aligned clip extraction. **60-70% review time reduction**, ~**85% alignment** with human-curated highlights. Demonstrates ML adaptation for domain-specific scientific content.

Driver Drowsiness Detection - Real-Time YOLOv8

Unified YOLOv8 detection; ~30% latency reduction, ~25% false positive reduction. Real-time OpenCV pipeline.

EDUCATION

Master of Science, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, SQL

LLM & AI: RAG, agentic workflows, LangChain, LlamaIndex, vector search, prompt engineering, structured outputs

ML: PyTorch, scikit-learn, XGBoost, HuggingFace Transformers, YOLOv8

Infra: Docker, FastAPI, Flask, Jenkins, Grafana, REST APIs

Work Authorization: F-1 OPT (currently US-authorized; open to EU work authorization discussions)